

Отчёт по лабораторной работе №7

Дисциплина: Администрирование сетевых подсистем

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Цель работы

Получить навыки настройки службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6.

Выполнение лабораторной работы

Настройка DHCP в случае IPv4

Запустил GNS3 VM и GNS3, создал новый проект. Изменил имена устройств согласно шаблону и включил захват трафика (рис. @fig:001):

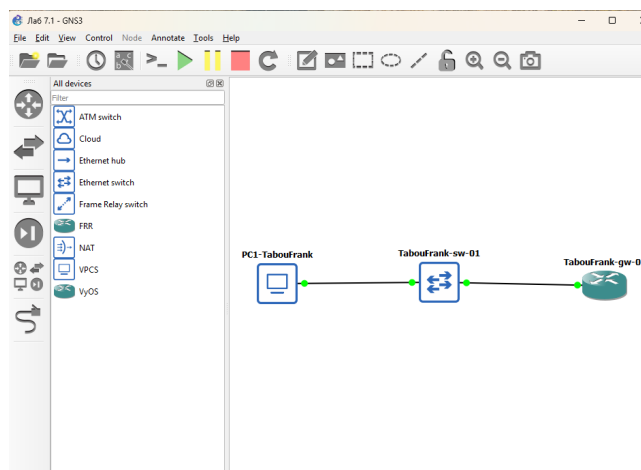


Figure 1: Топология сети с включённым захватом трафика

Настроил системные параметры VyOS: имя хоста `alkamal-gw-01`, домен `alkamal.net`, создал пользователя `alkamal` (рис. @fig:002):

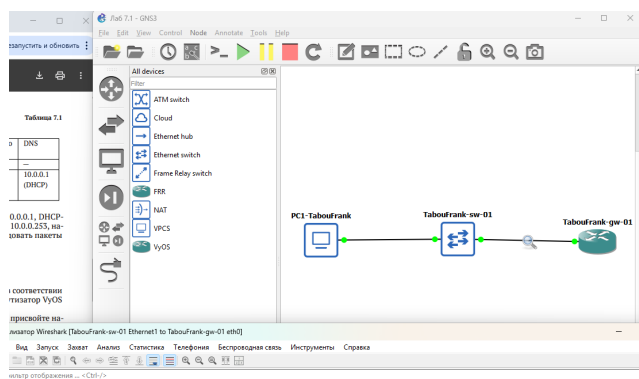


Figure 2: Настройка системных параметров VyOS

Выполнил повторный вход под новым пользователем и удалил учётную запись `vyos` (рис. @fig:003):

Назначил статический IPv4-адрес `10.0.0.1/24` интерфейсу `eth0` (рис. @fig:004):

Настроил DHCP-сервер на маршрутизаторе для сети `10.0.0.0/24` (рис. @fig:005):

Проверил статистику DHCP-сервера: активных аренд нет (рис. @fig:006):

На узле PC1 выполнил команду `ip dhcp -d` для получения адреса по DHCP (рис. @fig:007, @fig:008):

Проверил полученные параметры и связность с маршрутизатором (рис. @fig:009):

На маршрутизаторе проверил статистику и информацию о выданной аренде (рис. @fig:010):

Проанализировал журнал DHCP-сервера (рис. @fig:011):

В Wireshark зафиксирован полный процесс DHCP (рис. @fig:012, @fig:013):

```

Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ install image
You are trying to install from an already installed system. An ISO
image file to install or URL must be specified.
Exiting...
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name TabouFrank-gw-01
Configuration path: [system] is not valid
Set failed

[edit]
vyos@vyos# set system host-name TabouFrank-gw-01
[edit]
vyos@vyos# set system domain-name TabouFrank.net
[edit]
vyos@vyos# set system login user TabouFrank authentication plaintext-password
TabouFrank
Configuration path: system login user TabouFrank [authentication] is not val
id
Set failed

[edit]
vyos@vyos# set system login user TabouFrank authentication plaintext-password d
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$

```

Figure 3: Повторный вход и удаление учётной записи

```

Welcome to VyOS - TabouFrank-gw-01 ttyS0
TabouFrank-gw-01 login: TabouFrank
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
TabouFrank@TabouFrank-gw-01:~$

```

Figure 4: Назначение статического IPv4-адреса

```

Welcome to VyOS - TabouFrank-gw-01 ttyS0
TabouFrank-gw-01 login: TabouFrank
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
TabouFrank@TabouFrank-gw-01:~$ configure
[edit]
TabouFrank@TabouFrank-gw-01# delete system login user vyos
[edit]
TabouFrank@TabouFrank-gw-01# commit
[edit]
TabouFrank@TabouFrank-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
TabouFrank@TabouFrank-gw-01#

```

Figure 5: Конфигурация DHCP-сервера

```

Welcome to VyOS - TabouFrank-gw-01 tty50
TabouFrank-gw-01 login: TabouFrank
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
TabouFrank@TabouFrank-gw-01:~$ configure
[edit]
TabouFrank@TabouFrank-gw-01# delete system login user vyos
[edit]
TabouFrank@TabouFrank-gw-01# commit
[edit]
TabouFrank@TabouFrank-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
TabouFrank@TabouFrank-gw-01# set interfaces ethernet eth0 address 10.0.0.1/24
[edit]
mk domain-name TabouFrank.net

Configuration path: service dhcp-server [shared-network-nae] is not valid
Set failed

[edit]
mk domain-name TabouFrank.net et service dhcp-server shared-network-nae TabouFrank
mk domain-name TabouFrank.net et service dhcp-server shared-network-name TabouFrank
[edit]
mk name-server 10.0.0.1-01# set service dhcp-server shared-network-name TabouFrank
[edit]
mk subnet 10.0.0.0/24 default-router 10.0.0.1server shared-network-name TabouFrank
[edit]
mk subnet 10.0.0.0/24 range hosts start 10.0.0.2ver shared-network-name TabouFrank
[edit]
mk subnet 10.0.0.0/24 range hosts stop 10.0.0.253
[edit]
TabouFrank@TabouFrank-gw-01#

```

Figure 6: Статистика DHCP-сервера

```

[edit]
TabouFrank@TabouFrank-gw-01# set interfaces ethernet eth0 address 10.0.0.1/24

Configuration path: [interfaces ethernet eth0 address 10.0.0.1/24] already exists

[edit]
TabouFrank@TabouFrank-gw-01# save
Warning: you have uncommitted changes that will not be saved.

Saving configuration to '/config/config.boot'...
Done
[edit]
TabouFrank@TabouFrank-gw-01# commit

Can't configure both static IPv4 and DHCP address on the same interface

[[interfaces ethernet eth0]] failed
[ service dhcp-server ]
None of the configured subnets have an appropriate primary IP address on
any broadcast interface configured, nor was there an explicit listen-
address configured for serving DHCP relay packets!

[[service dhcp-server]] failed
Commit failed
[edit]
TabouFrank@TabouFrank-gw-01# address dhcp

Invalid command: [address]

[edit]
TabouFrank@TabouFrank-gw-01# show configuration commands | match eth0

Configuration path: [configuration] is not valid
Show failed

[edit]
TabouFrank@TabouFrank-gw-01# show configuration commands | match eth0

Configuration path: [configuration] is not valid
Show failed

[edit]
TabouFrank@TabouFrank-gw-01# delete interfaces ethernet eth0 address dhcp
[edit]
TabouFrank@TabouFrank-gw-01# set interfaces ethernet eth0 address 10.0.0.1/24

Configuration path: [interfaces ethernet eth0 address 10.0.0.1/24] already exists

[edit]
TabouFrank@TabouFrank-gw-01# show configuration commands | match eth0

Configuration path: [configuration] is not valid
Show failed

[edit]
TabouFrank@TabouFrank-gw-01# commit
[edit]
TabouFrank@TabouFrank-gw-01#

```

Figure 7: DHCP Discover и Offer

```

TabouFrank@TabouFrank-gw-01$ save
Saving configuration to /config/config.boot'...
Done
[edit]
TabouFrank@TabouFrank-gw-01$ exit
exit
TabouFrank@TabouFrank-gw-01$ show dhcp server statistics
-----
Pool      Size      Leases    Available  Usage
-----
TabouFrank 252      0         252      0%
TabouFrank@TabouFrank-gw-01$ show dhcp server leases
-----
IP address  Hardware address  State  Lease start  Lease expiration  Remaining  Pool  Hostname
-----
TabouFrank@TabouFrank-gw-01$

```

Figure 8: DHCP Request и ACK

```

TabouFrank@TabouFrank-gw-01$ show interfaces
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface  IP Address  S/L  Description
-----
eth0       10.0.0.1/24  --   u/u
eth1       -           --   u/D
eth2       -           --   u/D
ls         127.0.0.1/8  --   u/u
ls         ::1/128     --   u/u
TabouFrank@TabouFrank-gw-01$

```

Figure 9: Проверка параметров и связности

```

VPCS> ip dhcp -d
Opcode: 1 (REQUEST)
Client IP Address: 0.0.0.0
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Discover
Option 12: Host Name = VPCS
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:68:00

Opcode: 1 (REQUEST)
Client IP Address: 0.0.0.0
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Discover
Option 12: Host Name = VPCS
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:68:00

Opcode: 2 (REPLY)
Client IP Address: 0.0.0.0
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Offer
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = TabouFrank.net

Opcode: 1 (REQUEST)
Client IP Address: 10.0.0.2
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Request
Option 54: DHCP Server = 10.0.0.1

```

Figure 10: Статистика и информация об аренде

```

VPCS> save
Saving startup configuration to startup.vpc
VPCS>

VPCS> show ip
NAME                : VPCS[1]
IP/BRK              : 10.0.0.2/24
GATEWAY              : 10.0.0.1
DNS                  : 10.0.0.1
DHCP SERVER          : 10.0.0.1
DHCP LEASE           : 86264, 86400/43200/75600
DOMAIN NAME          : TabouFrank.net
MAC                  : 00:50:79:66:60:00
LPORT                : 10054
RHOST:PORT            : 127.0.0.1:10055
MTU                   : 1500

VPCS> ping 10.0.0.1 -c 2

64 bytes from 10.0.0.1 icmp_seq=1 ttl=64 time=3.137 ms
64 bytes from 10.0.0.1 icmp_seq=2 ttl=64 time=2.496 ms

VPCS>

```

Figure 11: Журнал DHCP-сервера

```

TabouFrank@TabouFrank-pc-01:~$ show dhcp server statistics
Pool
-----
TabouFrank      252      1      251 OK
TabouFrank@TabouFrank-pc-01:~$ show dhcp server statistics
Pool
-----
TabouFrank      252      1      251 OK
TabouFrank@TabouFrank-pc-01:~$ show dhcp server leases
IP address      Hardware address  State  Lease start
-----
10.0.0.2        00:50:79:66:60:00  active  2024/01/03 23:50:18  2026/01/04 23:50:18  23:51:55  TabouFrank VPCS
TabouFrank@TabouFrank-pc-01:~$

```

Figure 12: Захваченный DHCP-трафик

Wireshark packet capture analysis of DHCP traffic. The packet list shows a DHCP Discover (0) from 10.0.0.2 to 255.255.255.255. The packet details pane shows the DHCP Discover structure, including the Magic cookie (0x63325343), Transaction ID (0x641de55), and various options like Server host name (TabouFrank) and Host file name (not given).

Packet 1: DHCP Discover (0) from 10.0.0.2 to 255.255.255.255. Details: Magic cookie: 0x63325343, Transaction ID: 0x641de55, Server host name: TabouFrank, Host file name: not given.

Figure 13: Детализация DHCP-пакетов

Настройка DHCP в случае IPv6

Использовал Alpine Linux для поддержки DHCPv6. Топология сети с включённым захватом трафика (рис. @fig:014):

```
Taboufrank@Taboufrank-pc-01:~$ show log | grep dhcp
Jan 03 22:41:51 dhclient-script-vyos[1846]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 22:41:52 vyos-hosted[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:41:52 dhclient-script-vyos[1846]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 22:41:52 vyos-hosted[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:43:47 dhclient-script-vyos[1846]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 22:43:47 vyos-hosted[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:43:47 dhclient-script-vyos[1846]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 22:43:47 vyos-hosted[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:45:55 dhclient-script-vyos[2139]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 22:45:55 vyos-hosted[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:45:55 dhclient-script-vyos[2139]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 22:45:55 vyos-hosted[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:47:42 dhclient-script-vyos[2139]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 22:47:42 vyos-hosted[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:47:42 dhclient-script-vyos[2139]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 22:47:42 vyos-hosted[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:49:20 dhclient-script-vyos[2221]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 22:49:20 vyos-hosted[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:49:20 dhclient-script-vyos[2221]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 22:49:20 vyos-hosted[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:50:59 dhclient-script-vyos[2271]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 22:50:59 vyos-hosted[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:50:59 dhclient-script-vyos[2271]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 22:50:59 vyos-hosted[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:52:23 dhclient-script-vyos[2320]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 22:52:23 vyos-hosted[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:52:23 dhclient-script-vyos[2320]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 22:52:23 vyos-hosted[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:55:47 dhclient-script-vyos[2371]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 22:55:47 vyos-hosted[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:55:47 dhclient-script-vyos[2371]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 22:55:47 vyos-hosted[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:58:05 dhclient-script-vyos[2427]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 22:58:05 vyos-hosted[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:58:05 dhclient-script-vyos[2427]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 22:58:05 vyos-hosted[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:00:20 dhclient-script-vyos[2489]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 23:00:20 vyos-hosted[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:00:20 dhclient-script-vyos[2489]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 23:00:20 vyos-hosted[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:01:57 user@libexec/vyos/conf_mode_dhcp_server: /usr/libexec/vyos/conf_mode_dhcp_server
Jan 03 23:02:07 dhclient-script-vyos[2646]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 23:02:07 vyos-hosted[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:02:07 dhclient-script-vyos[2646]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 23:02:07 vyos-hosted[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:03:53 dhclient-script-vyos[2689]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 23:03:53 vyos-hosted[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:03:53 dhclient-script-vyos[2689]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 23:03:53 vyos-hosted[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:04:23 dhclient-script-vyos[2718]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 23:04:23 vyos-hosted[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:04:23 dhclient-script-vyos[2718]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Jan 03 23:04:23 vyos-hosted[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
```

Figure 14: Топология сети IPv6

Настроил системные параметры VyOS для IPv6 (рис. @fig:015):

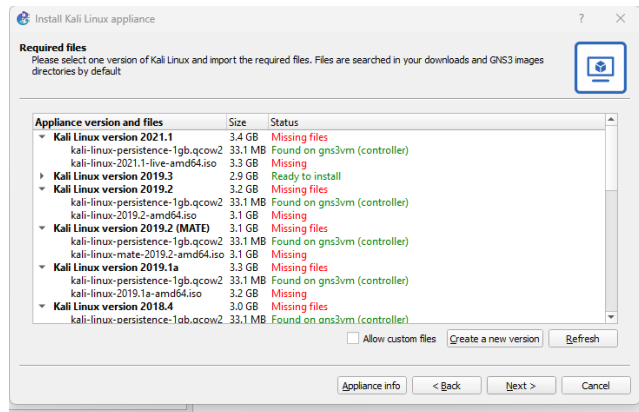


Figure 15: Настройка системных параметров VyOS

Назначил IPv6-адреса интерфейсам eth1 (2000::1/64) и eth2 (2001::1/64) (рис. @fig:016):

Настроил Stateless DHCPv6 и Router Advertisement на интерфейсе eth1 (рис. @fig:017, @fig:018):

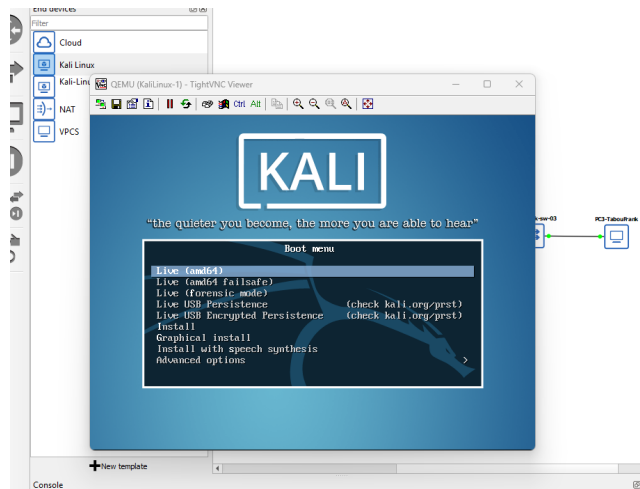


Figure 16: Назначение IPv6-адресов интерфейсам



Figure 17: Настройка Stateless DHCPv6

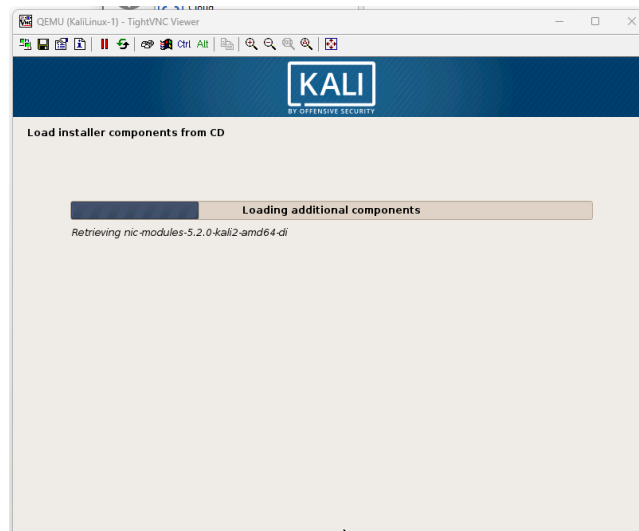


Figure 18: Просмотр конфигурации DHCPv6

Проверил IPv6-адресацию на узле PC2: адрес получен через SLAAC (рис. @fig:019):

Попытка получения адреса по DHCPv6 показала отсутствие опции IAADDR (рис. @fig:020):

Связность с маршрутизатором сохраняется (рис. @fig:021):

В захваченном трафике видна работа ICMPv6 и DHCPv6 Stateless (рис. @fig:022):

Настроил Stateful DHCPv6 на интерфейсе eth2 (рис. @fig:023):

Проверил состояние DHCPv6-сервера до запроса клиента (рис. @fig:024):

На узле PC3 выполнил получение адреса по Stateful DHCPv6 (рис. @fig:025):

Проверил IPv6-адресацию на узле PC3 после DHCPv6 (рис. @fig:026):

На маршрутизаторе появилась информация о выданной IPv6-аренде (рис. @fig:027):

В захваченном трафике зафиксирован полный процесс DHCPv6 Stateful (рис. @fig:028):

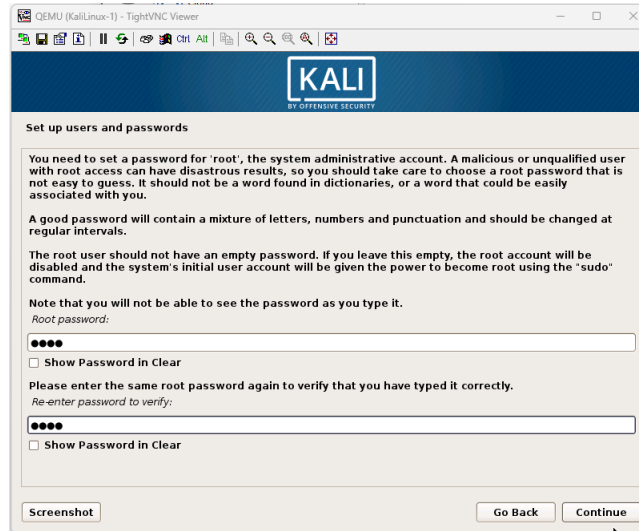


Figure 19: Проверка IPv6 на PC2

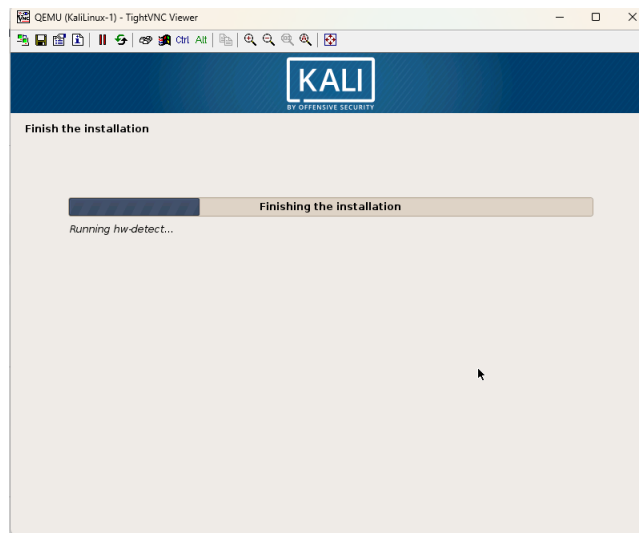


Figure 20: Попытка DHCPv6 на PC2

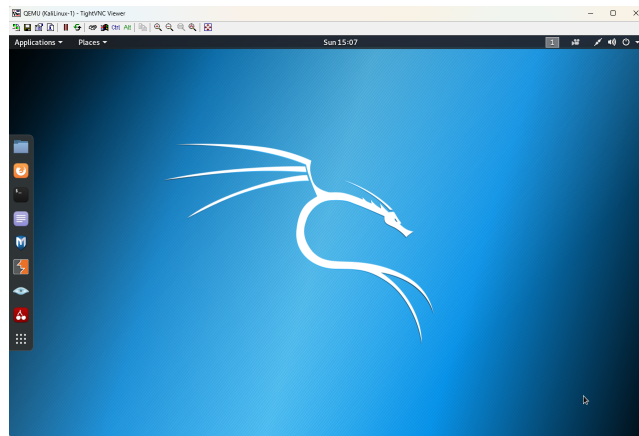


Figure 21: Проверка связности IPv6

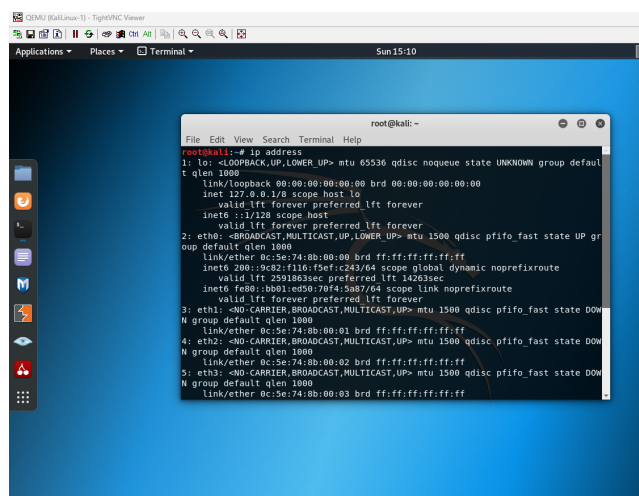


Figure 22: Трафик ICMPv6 и DHCPv6 Stateless



Figure 23: Настройка Stateful DHCPv6

```
root@kali:~# ping 2000::1 -c 2
PING 2000::1(2000::1) 56 data bytes

--- 2000::1 ping statistics ---
2 packets transmitted, 0 received, 100% packet loss, time 1025ms

root@kali:~# ping 2000::1 -c 2
PING 2000::1(2000::1) 56 data bytes

--- 2000::1 ping statistics ---
2 packets transmitted, 0 received, 100% packet loss, time 1005ms
```

Figure 24: Статистика DHCPv6

```
TabouFrank@TabouFrank-gw-01:~$ configure
[edit]
TabouFrank@TabouFrank-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication      Lease expiration      Remaining
Type      Pool      IAID_DUID
-----
[edit]
TabouFrank@TabouFrank-gw-01#
```

Figure 25: Получение адреса по DHCPv6

Filtered output [TabouFrank-gw-01 Ethernet to TabouFrank-gw-01 eth0]					
ID	Time	Source	Destination	Protocol	Length Info
268	4823.727860	0c:11:84:00:00:00	Broadcast	ARP	60 who has 10.0.0.2? Tell 10.0.0.1
269	4824.747547	0c:11:84:00:00:00	Broadcast	ARP	60 who has 10.0.0.2? Tell 10.0.0.1
270	4826.603940	0.0.0.0	255.255.255.255	DHCP	480 DHCP Request - Transaction ID 8bdc5d452
271	4826.790541	10.0.0.1	10.0.0.2	DHCP	342 DHCP ACK - Transaction ID 8bdc5d452
272	4827.683870	Private_06:00:00	Broadcast	ARP	64 Gratuitous ARP for 10.0.0.2 (request)
273	4828.655983	Private_06:00:00	Broadcast	ARP	64 Gratuitous ARP for 10.0.0.2 (request)
274	4829.685331	Private_06:00:00	Broadcast	ARP	64 Gratuitous ARP for 10.0.0.2 (request)
275	4894.422953	Private_06:00:00	Broadcast	ARP	60 who has 10.0.0.1? Tell 10.0.0.2
276	4894.422987	0c:11:84:00:00:00	Private_06:00:00	ARP	60 10.0.0.1 is at 0c:11:84:00:00:00
277	4894.434218	10.0.0.2	10.0.0.1	ICMP	90 Echo (ping) request (id=634b, seq=1256, ttl=64 (reply in 270))
278	4894.434682	10.0.0.1	10.0.0.2	ICMP	90 Echo (ping) reply (id=634b, seq=1256, ttl=64 (request in 277))
279	4895.437517	10.0.0.2	10.0.0.1	ICMP	90 Echo (ping) request (id=634b, seq=2712, ttl=64 (reply in 280))
280	4895.439464	10.0.0.1	10.0.0.2	ICMP	90 Echo (ping) reply (id=634b, seq=2712, ttl=64 (request in 279))
281	4899.322120	0c:11:84:00:00:00	Private_06:00:00	ARP	60 who has 10.0.0.2? Tell 10.0.0.1
282	4899.322580	Private_06:00:00	0c:11:84:00:00:00	ARP	60 10.0.0.2 is at 0c:11:84:00:00:00
283	5262.270844	Private_06:00:00	Broadcast	ARP	64 who has 10.0.0.1? Tell 10.0.0.2
284	5262.270112	0c:11:84:00:00:00	Private_06:00:00	ARP	60 10.0.0.1 is at 0c:11:84:00:00:00
285	5262.279527	10.0.0.2	10.0.0.1	ICMP	90 Echo (ping) request (id=634b, seq=1256, ttl=64 (reply in 286))
286	5262.281146	10.0.0.1	10.0.0.2	ICMP	90 Echo (ping) reply (id=634b, seq=1256, ttl=64 (request in 285))
287	5263.281870	10.0.0.2	10.0.0.1	ICMP	90 Echo (ping) request (id=634b, seq=2712, ttl=64 (reply in 288))
288	5263.282793	10.0.0.1	10.0.0.2	ICMP	90 Echo (ping) reply (id=634b, seq=2712, ttl=64 (request in 287))
289	5266.220806	10.0.0.2	10.0.0.1	ICMP	90 Echo (ping) request (id=634b, seq=1256, ttl=64 (reply in 289))
290	5266.221580	10.0.0.1	10.0.0.2	ICMP	90 Echo (ping) reply (id=634b, seq=1256, ttl=64 (request in 289))
291	5267.244880	10.0.0.2	10.0.0.1	ICMP	90 Echo (ping) request (id=634b, seq=2712, ttl=64 (reply in 292))
292	5267.226655	10.0.0.1	10.0.0.2	ICMP	90 Echo (ping) reply (id=634b, seq=2712, ttl=64 (request in 291))
293	5267.780640	0c:11:84:00:00:00	Private_06:00:00	ARP	60 who has 10.0.0.2? Tell 10.0.0.1
294	5267.787352	Private_06:00:00	0c:11:84:00:00:00	ARP	60 10.0.0.2 is at 0c:11:84:00:00:00

Figure 26: Проверка IPv6 на PC3

```
TabouFrank-gw-01 - PuTTY
Welcome to VyOS - TabouFrank-gw-01 ttyS0

TabouFrank-gw-01 login: TabouFrank
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
TabouFrank@TabouFrank-gw-01:~$ configure
[edit]
TabouFrank@TabouFrank-gw-01# set service router-advert interface eth2 managed-fl
g
[edit]
TabouFrank@TabouFrank-gw-01# set service dhcpv6-server shared-network-name Tabou
Frank-stateful
[edit]
TabouFrank@TabouFrank-gw-01# set service dhcpv6-server shared-network-name Tabou
Frank-stateful subnet 2001::0/64
[edit]
TabouFrank@TabouFrank-gw-01# set service dhcpv6-server shared-network-name Tabou
Frank-stateful subnet 2001::0/64 name-server 2001::1
[edit]
TabouFrank@TabouFrank-gw-01# set service dhcpv6-server shared-network-name Tabou
Frank-stateful subnet 2001::0/64 domain-search TabouFrank.net
[edit]
TabouFrank@TabouFrank-gw-01# set service dhcpv6-server shared-network-name Tabou
Frank-stateful subnet 2001::0/64 address-range start 2001::100 stop 2001::199
[edit]
TabouFrank@TabouFrank-gw-01# commit
[edit]
TabouFrank@TabouFrank-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
TabouFrank@TabouFrank-gw-01#
```

Figure 27: Информация о выданной аренде

```
TabouFrank@TabouFrank-gw-01# run show dhcpv6 server leases
IPv6 address  State  Last communication  Lease expiration  Remaining  Type  Pool  IAID
-----
[edit]
TabouFrank@TabouFrank-gw-01#
```

Figure 28: Трафик DHCPv6 Stateful

Выводы

В ходе выполнения лабораторной работы была успешно реализована и исследована настройка автоматической адресации IPv4 и IPv6 с использованием протоколов DHCP и DHCPv6 в двух режимах: без отслеживания состояния (stateless) и с отслеживанием состояния (stateful).

На интерфейсе eth1 маршрутизатора был настроен режим DHCPv6 Stateless, при котором узел PC2 получил IPv6-адрес с помощью механизма SLAAC, а дополнительные параметры сети были получены по протоколу DHCPv6.

На интерфейсе eth2 маршрутизатора был настроен режим DHCPv6 Stateful, при котором узел PC3 получил IPv6-адрес из заданного диапазона, а также параметры DNS от DHCPv6-сервера.

Анализ захваченного сетевого трафика показал корректную работу протоколов ICMPv6, Router Advertisement, Neighbor Discovery и DHCPv6, а также последовательность обмена сообщениями при получении адреса.

Таким образом, все поставленные в задании цели были достигнуты: выполнена настройка IPv4/IPv6-адресации, реализованы режимы DHCP, подтверждена их корректная работа и проанализировано взаимодействие сетевых устройств на уровне протоколов.